

Pipeline

(Sakshi)
property.py

- combines raw data into one dataframe
- filters data to Christchurch city
- adds month-year column

output:

- categorical/numerical variables summary
- missing values summary

christchurch-data.csv

(William)

airbnb-christchurch-prices.py

- display data about prices into histogram

output:

- display data about prices into histogram

(Đào)

calculate how long last review

- Make a new column called "days since last review"
- Print out the missing values and days since last scraped
- Display data about days since last review

output:

- displays data about days since last review into a histogram

(Trần)

filter-chch-top-ten-reviews.py

- filter data to 4 columns
- rearrange data by top reviews

output:

- top reviewed airbnb.csv
- total number of Christchurch Airbnbs

christchurch-top-ten-filtered.csv

(Đào)

deliverable-week8.py

- keep columns: id, latitude, longitude, neighborhood, room type, price, availability365, month_year, minimum_nights

christchurch-airbnb-cleaned.csv



(William)
tenancy-data-cleaning.py

- Filter tenancy data to 2025 October - 2026 April (keep date as string)
- filter columns: timeframe, Location-Id, Dwelling type, Number of Beds, Median Rent, Total Bonds
- remove all na values
- filter to summary statistics

Detailed-Quarterly-Tenancy.csv

(Trần)
append-ssa2026-code-name.py

- get the ssa2026 code and names for the coordinates for AirBnb

christchurch-air-bnb-code-name-appended.csv

Output:

- Print the locations of top 20 Airbnb median price gaps
- print the top 15 locations boxplot
- compare short-term and long-term rental prices

(Đào)
gap-between-short-term-and-long-term.py

- reclassify airbnbs into short term and long term airbnbs
- calculate the price gaps between long term and short term airbnbs

(William)

comparison-by-region.py

- find the total Airbnbs and total bonds for each sa2026 area code
- produce a viewable html interactive map

christchurch-airbnb-tenancy-joined.csv

Output:

- print the median price
- print out the summary for missing values

(Sakshi)
combine-data.py

- Convert the month_year and TimeFrame columns to just contain just month
- Join Airbnb data and Tenancy data using left join
- calculate median price
- check merge results

Output:

Viewable html interactive map